Solute transport

SWAT+ Documentation

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Contents

Purpose . . . 1
Activation . . . 2
File overview . . . 2
Missing files . . . 2
salt_hru.ini . . . 3
salt_aqu.ini . . . 3
salt_channel.ini . . . 3
salt_res . . . 3
salt_irrigation . . . 4
salt_fertilizer.frt . . . 4
salt_road . . . 4
salt_plants . . . 5
salt_uptake . . . 5
salt_urban . . . 5
Example . . . 6
Related files . . . 6
Purpose . . . 7
constituents.cs . . . 7
File overview . . . 7
Missing files . . . 8
cs_hru.ini . . . 8
cs_aqu.ini . . . 9
cs_channel.ini . . . 9
cs_res . . . 9
cs_irrigation . . . 10
fertilizer.frt_cs . . . 10
cs_reactions . . . 10
cs_uptake . . . 11
cs_plants_boron . . . 11
cs_urban . . . 11
cs_streamobs . . . 11
Example . . . 11
Related files . . . 12

Purpose

The salt module tracks eight salt ions in soil water, groundwater, channels, reservoirs and wetlands, together with five salt minerals in soils and aquifers that dissolve and precipitate. Salt enters from initial conditions, fertilizer and manure, road salt, atmospheric deposition and point sources, and leaves with plant uptake, runoff, percolation and flow. High soil salinity can reduce plant growth.

The salt files are not listed in `file.cio`; SWAT+ looks for them by name in the run directory. The constituent module, which uses the same master file, is on the constituent inputs page.

Activation

Salt simulation is switched on by listing salt ions in the salt block of `constituents.cs`. With no salt ions listed, or without `constituents.cs`, the salt files are ignored.

The ions must be listed in this order, because several routines use the position of an ion and not its name:

Position	Ion
1	`so4` (sulfate)
2	`ca` (calcium)
3	`mg` (magnesium)
4	`na` (sodium)
5	`k` (potassium)
6	`cl` (chloride)
7	`co3` (carbonate)
8	`hco3` (bicarbonate)

Every file below that has one value per ion gives the values in this order. Files with mineral fractions give five values in this order: $CaCO_3$, $MgCO_3$, $CaSO_4$, $MgSO_4$, NaCl.

File overview

File	Content	Linked to objects by
`salt_hru.ini`	Initial ion concentrations and mineral fractions in HRU soils	`salt` column of `soil_plant.ini`
`salt_aqu.ini`	Initial ion concentrations and mineral fractions in aquifers	`salt` column of `initial.aqu_cs`
`salt_channel.ini`	Initial ion concentrations in channels	`salt` column of `initial.cha_cs`
`salt_res`	Initial ion concentrations in reservoirs and wetlands	`salt` column of `reservoir.res_cs` and `wetland.wet_cs`
`salt_irrigation`	Ion concentrations in irrigation water from outside the model	position, see below
`salt_fertilizer.frt`	Ion content of fertilizers and manures	position in the fertilizer list
`salt_road`	Road salt applied to the land surface	atmospheric deposition stations
`salt_atmo.cli`	Ion concentration in rainfall and dry deposition	atmospheric deposition stations, see climate inputs
`salt_recall.rec`	Ion loads from point sources	see recall inputs
`salt_plants`	Salt tolerance of each plant	position in `plants.plt`
`salt_uptake`	Daily ion uptake by each plant	position in `plants.plt`
`salt_urban`	Ion content of sediment from impervious areas	urban type name in `urban.urb`

Missing files

`salt_hru.ini`, `salt_aqu.ini` and `salt_channel.ini` are opened whether or not they exist. If one is missing, SWAT+ creates an empty file of that name in the run directory and reads no records from it. An empty file, whether created this way or supplied, is read the same way: SWAT+ finds no title line and no records.

The test projects `refdata/Ames_sub1` and `refdata/Osu_1hru` in the SWAT+ repository contain empty `salt_hru.ini`, `salt_aqu.ini` and `salt_channel.ini` files created by earlier runs. Both projects give `null` for the constituent file in `file.cio`, so no salt ions are simulated and the empty files have no effect on the results. If salt ions are listed in `constituents.cs`, an empty file behaves as follows:

- Empty `salt_hru.ini`: there is no record for the HRUs, but HRU setup still reads record 1, which does not exist. Give at least one record.

- Empty `salt_aqu.ini`: no aquifer finds a record, so aquifers start with no salt ions and no salt minerals.
- Empty `salt_channel.ini`: there is no record for the channels, but channel setup still reads record 1, which does not exist. Give at least one record.

Some files are needed as soon as salt ions are listed, because the setup reads from them for every object without checking that they were found:

- `salt_irrigation` is read for every HRU during setup.
- `salt_road` is read every day for every HRU when `atmodep.cli` is in use.

Provide these files whenever salts are simulated, with zero values if there is nothing to add.

salt_hru.ini

Initial salt in HRU soils. Lines 1 to 5 are skipped. Each record then takes three lines:

Line of record	Content
1	Record name
2	Soil values: 8 ion concentrations in soil water (mg/L), then 5 mineral fractions (percent)
3	Plant values: 13 numbers, read but not used

An HRU uses the record named in the `salt` column of its `soil_plant.ini` record. If the name is not found, the first record is used. Every soil layer starts with the same values. The ion mass in a layer is the concentration times the water stored in the layer at the start of the run.

salt_aqu.ini

Initial salt in aquifers. Lines 1 to 4 are skipped. Each later line is one record: name, 8 ion concentrations in groundwater (mg/L), and 5 mineral fractions (percent).

Aquifers find their record through `initial.aqu_cs`, which has a title line, a header line and one row per aquifer with the columns `name`, `pest`, `path`, `hmet`, `salt` and `cs`. The `salt` column names a record in `salt_aqu.ini`. For salts, aquifer n uses row n of `initial.aqu_cs`, whatever the row's name, so the rows must be in aquifer order. The ion mass is the concentration times the groundwater stored at the start of the run.

salt_channel.ini

Initial salt in channel water. Lines 1 and 2 are skipped. Each later line is one record: name and 8 ion concentrations (mg/L).

Channels find their record through `initial.cha_cs` (title, header, then rows with `name`, `pest`, `path`, `hmet`, `salt`, `cs`). A channel uses the `initial.cha_cs` row at the same position as the `initial.cha` row it selects by name; the `salt` column of that row names a record in `salt_channel.ini`. If the name is not found, the first record is used.

salt_res

Initial salt in reservoirs and wetlands. Lines 1 to 11 are skipped (two title lines, eight comment lines and a header). Each later line is one record: name and 8 ion concentrations (mg/L). The file is ignored when the reservoir nutrient file in `file.cio` is `null`.

Reservoirs find their record through `reservoir.res_cs` and wetlands through `wetland.wet_cs`. Both have two header lines and one row per reservoir (or wetland) data record with the columns `id`, `pest`, `weir`, `salt` and `cs`. `id` is the position of the record in `reservoir.res` or `wetland.wet`, and `salt` names a record in `salt_res`. A wetland without a matching record starts with no salt.

salt_irrigation

Ion concentrations in irrigation water taken from outside the model. Lines 1 and 2 are skipped. Each later line is one record: name and 8 ion concentrations (mg/L).

An HRU uses the record at the same position as the `salt_hru.ini` record it uses (record 3 of `salt_hru.ini` goes with record 3 of `salt_irrigation`); the name is not used. In 62.0.1 the transfer of salt with irrigation water is switched off in water allocation, so these concentrations do not change the salt balance.

salt_fertilizer.frt

Ion content of each fertilizer. Line 1 is a title and line 2 a header. Then one row per record of `fertilizer.frt`, in the same order and with the same number of rows:

#	Column	Units	Description
1	`name`	-	Fertilizer name. Not checked against `fertilizer.frt`.
2	`so4`	kg/kg	Sulfate per kg of material applied.
3	`ca`	kg/kg	Calcium per kg of material applied.
4	`mg`	kg/kg	Magnesium per kg of material applied.
5	`na`	kg/kg	Sodium per kg of material applied.
6	`k`	kg/kg	Potassium per kg of material applied.
7	`cl`	kg/kg	Chloride per kg of material applied.
8	`co3`	kg/kg	Carbonate per kg of material applied.
9	`hco3`	kg/kg	Bicarbonate per kg of material applied.

The salt added (kg/ha) is the amount applied (kg/ha) times the value in the table.

Where salt is added in 62.0.1:

- A manure operation in a management schedule or a `manure` action in a decision table adds salt, split between the top two soil layers by the surface fraction of the application method in `chem_app.ops`. The row used is the row at the position of the manure in `manure_db.frt`, and salt is added only when the manure's `org_min` entry matches a record in `manure_om.frt` other than the first.
- A fertilizer operation in a management schedule adds salt only when the HRU has standing water, and the salt goes into the ponded water.
- A fertilizer operation on dry soil adds no salt.

Salt from a fertilizer whose name in `fertilizer.frt` starts with `a` is reported as a soil amendment in the salt balance output; other fertilizers are reported as fertilizer.

salt_road

Road salt applied to the soil surface. The file is organised by the atmospheric deposition stations of `atmodep.cli` and uses the same time step as that file. Lines 1 to 4 are skipped. Then, for each station in the order of `atmodep.cli`, one line with the station name followed by one line per ion (8 lines), each starting with the ion name:

`atmodep.cli` time step	Values on each ion line	Daily application
`aa`	1 value, kg/ha per year	the value / 365, every day
`mo`	1 value per month of the deposition record, kg/ha per month	the month's value / days in the month
`yr`	1 value per year of the deposition record, kg/ha per year	spread over the days with precipitation and a mean temperature below 0 °C, in proportion to the day's precipitation

Each HRU receives the road salt of the deposition station of its weather station, on its whole area, into the top soil layer. For the `yr` time step the precipitation and temperature used to spread the salt come from the weather station at the same position in the station list as the deposition station.

salt_plants

Salt tolerance of plants and switches for salt stress.

Line	Content
1, 2	Title and header, skipped
3	Conversion factor from total dissolved solids (mg/L) to electrical conductivity (dS/m)
4	Skipped
5	Salt stress switch: 0 = off, 1 = on
6	Soil type: 1 = gypsiferous ($CaSO_4$) soils, 2 = NaCl soils
7	Stress method: 1 = salt stress applied after the other stresses, 2 = salt stress included with the other stresses
8 to 11	Skipped
12	Header, skipped
13+	One row per plant in `plants.plt`, in the same order: name, threshold a (dS/m), slope b (% per dS/m)

The plant name is not checked; the row position decides.

Salt stress

When the switch is on, SWAT+ computes for each soil layer within the root zone the total dissolved solids of the soil water at saturation, after mineral dissolution, and converts it to the electrical conductivity of the saturation extract EC_e with the factor on line 3. The relative yield for that layer is

$$Y_r = \frac{100 - b\,(EC_e - a)}{100} \qquad \text{for } EC_e > a$$

limited to 0 to 1, and 1 when $EC_e \le a$. For gypsiferous soils 2 dS/m is added to a. A plant with $a = 0$ has no salt stress. The salt stress of the plant is the smallest Y_r over its root zone layers.

With stress method 1, plant growth is multiplied by the salt stress after the smallest of the water, temperature, nitrogen, phosphorus and aeration stresses has been applied. With method 2, the salt stress is one of the stresses and the smallest of them is used.

salt_uptake

Specified daily salt uptake by plants. Lines 1 to 3 are skipped. Then one row per plant in `plants.plt`, in the same order: name, and 8 uptake values (kg/ha per day). The plant name is not checked.

While a plant has roots, its daily uptake of each ion is split between the soil layers in proportion to the root mass in each layer and is limited to the ion mass in the layer. Without the file, plants take up no salt.

salt_urban

Ion content of sediment washed from impervious areas. Lines 1 and 2 are skipped. Each later line is an urban type name from `urban.urb` and 8 concentrations (mg of ion per kg of sediment). In 62.0.1 the values are read but not used.

Example

`salt_plants` in the layout SWAT+ Editor writes, with salt stress on and illustrative values:

```
salt_plants: written by SWAT+ editor
conversion factor from TDS --> EC (EC=TDS/factor)
640.0
plant salinity tolerance
1            flag (0 = off; 1 = on)
2            flag (1=caso4 soils; 2=nacl soils)
1            flag (1=salt stress applied after other stresses applied; 2=included with other stresses)
NOTE: a and b values are for nacl soils
NOTE: if a and b values = 0, no salt impact
a = EC_sat threshold for impact on crop yield
b = slope of salinity impact on crop yield
            name           a           b
            agrl        1.70       12.00
            corn        1.70       12.00
```

TDS is divided by 640 to give EC_e. For corn, with $EC_e = 4$ dS/m in a layer, the relative yield is $(100 - 12 \times (4 - 1.7))/100 = 0.72$. The rows after the header continue for every plant in `plants.plt`.

Related files

- `constituents.cs`: lists the salt ions and switches the salt module on.
- `soil_plant.ini`: names the `salt_hru.ini` record for each HRU.
- `fertilizer.frt`: sets the row order of `salt_fertilizer.frt`.
- `atmodep.cli`, `salt_atmo.cli`: deposition stations and salt deposition.
- `salt_recall.rec`: salt loads from point sources.
- gwflow inputs: salt ions become groundwater solutes when gwflow solute transport is on.
- constituent inputs: the selenium and boron module.

Sources

- Master list: `constit_db_read.f90`
- Readers: `salt_hru_read.f90`, `salt_aqu_read.f90`, `salt_cha_read.f90`, `res_read_saltdb.f90`, `salt_irr_read.f90`, `salt_fert_read.f90`, `salt_roadsalt_read.f90`, `salt_plant_read.f90`, `salt_uptake_read.f90`, `salt_urban_read.f90`
- Link files: `aqu_read_init_cs.f90` (`initial.aqu_cs`), `ch_read_init_cs.f90` (`initial.cha_cs`) with `sd_channel_read.f90`, `res_read_salt_cs.f90`, `wet_read_salt_cs.f90`, `soil_plant_init.f90` and `hru_read.f90`
- Initialisation: `salt_hru_init.f90`, `sd_hydsed_init.f90`, `res_initial.f90`, `wet_initial.f90`
- Processes: `salt_fert.f90`, `salt_fert_wet.f90` (called from `mgt_sched.f90` and `actions.f90`), `salt_irrig.f90`, `salt_roadsalt.f90`, `salt_uptake.f90`, `salt_chem_soil_single.f90`, `salt_chem_aqu.f90`, salt stress in `pl_waterup.f90` and `pl_biomass_gro.f90`
- Types and variables: `constituent_mass_module.f90`, `salt_module.f90`, `salt_data_module.f90`, `res_salt_module.f90`

File and column	Fortran variable
`salt_hru.ini` soil and plant lines	`salt_soil_ini%soil`, `salt_soil_ini%plt`
`salt_aqu.ini`	`salt_aqu_ini%conc`, `salt_aqu_ini%frac`
`salt_channel.ini`	`salt_cha_ini%conc`
`salt_res`	`res_salt_data%c_init`
`salt_irrigation`	`salt_water_irr%water`, copied to `cs_irr%saltc`
`salt_fertilizer.frt`	`fert_salt` (type `fert_db_salt`)
`salt_road`	`rdapp_salt%salt%road`, `%roadmo`, `%roadyr`, `%roadday`
`salt_plants` lines 3, 5, 6, 7	`salt_tds_ec`, `salt_tol_sim`, `salt_soil_type`, `salt_effect`
`salt_plants` columns a, b	`salt_stress_a`, `salt_stress_b`

File and column	Fortran variable
salt_uptake	salt_uptake_kg
salt_urban	salt_urban_conc
initial.aqu_cs	aqu_init_dat_c_cs
initial.cha_cs	ch_init_cs
reservoir.res_cs, wetland.wet_cs	res_dat_c_cs, wet_dat_c_cs

Purpose

constituents.cs lists the pesticides, pathogens, heavy metals, salt ions and other constituents that a run simulates. The last group, the other constituents, is the constituent module: selenate (SeO_4), selenite (SeO_3) and boron in soils, aquifers, channels, reservoirs and wetlands. The module simulates sorption, selenium reduction and selenium release from shale, and the constituents move with water like the other dissolved substances.

The salt ion files are described on the salt inputs page. Apart from constituents.cs, which is named in file.cio, the files are not listed in file.cio; SWAT+ looks for them by name in the run directory.

constituents.cs

The master list, named on the simulation line of file.cio (default constituents.cs). If the file is missing or named null, no pesticides, pathogens, salts or other constituents are simulated.

Line 1 is a title. Then come five groups, each as a count followed by that many names:

Group	Count	Names
1	number of pesticides	pesticide names, matched by name to pesticide.pes
2	number of pathogens	pathogen names, matched by name to pathogens.pth
3	number of heavy metals	metal names, not matched to any database (see metals)
4	number of salt ions	salt ion names, in the order given on the salt inputs page
5	number of other constituents	constituent names

Each count is read from a separate line and the names that follow are read as one list, starting on the next line, so they can be on one line or several. A count of 0 still needs its names line: SWAT+ skips one line after the count, so leave a blank line (or any text) there. A pesticide or pathogen name that is not in its database is kept in the list but gets no parameters.

The constituent module uses the position of each constituent and not its name. Fertilizer input, selenium reactions and the reservoir parameters assume this order:

Position	Constituent
1	seo4 (selenate)
2	seo3 (selenite)
3	boron

File overview

File	Content	Linked to objects by
cs_hru.ini	Initial dissolved and sorbed constituents in HRU soils	cs column of soil_plant.ini

File	Content	Linked to objects by
cs_aqu.ini	Initial dissolved and sorbed constituents in aquifers	cs column of initial.aqu_cs
cs_channel.ini	Initial concentrations in channels	cs column of initial.cha_cs
cs_res	Reservoir and wetland parameters and initial concentrations	cs column of reservoir.res_cs and wetland.wet_cs
cs_irrigation	Concentrations in irrigation water from outside the model	position, see below
fertilizer.frt_cs	Constituent content of fertilizers and manures	position in the fertilizer list
cs_reactions	Reaction parameters for soils and aquifers	HRU and aquifer order
cs_uptake	Daily constituent uptake by each plant	position in plants.plt
cs_atmo.cli	Concentration in rainfall and dry deposition	atmospheric deposition stations, see climate inputs
cs_recall.rec	Loads from point sources	see recall inputs
cs_plants_boron	Boron tolerance of plants	read but not used
cs_urban	Constituent content of sediment from impervious areas	read but not used
cs_streamobs	Channels for daily concentration output	not read

Missing files

cs_hru.ini, cs_aqu.ini and cs_channel.ini are opened whether or not they exist. If one is missing, SWAT+ creates an empty file of that name in the run directory and reads no records from it. An empty file, whether created this way or supplied, is read the same way.

The test project refdata/Ames_sub1 in the SWAT+ repository contains empty cs_hru.ini, cs_aqu.ini and cs_channel.ini files created by earlier runs. It gives null for the constituent file in file.cio, so no constituents are simulated and the empty files have no effect on the results. If constituents are listed in constituents.cs, an empty file behaves as follows:

- Empty cs_hru.ini: there is no record for the HRUs, but HRU setup still reads record 1, which does not exist. Give at least one record.
- Empty cs_aqu.ini: when initial.aqu_cs is present, the aquifer search runs over the number of records in cs_irrigation and reads cs_aqu.ini records that do not exist. Give at least as many records as cs_irrigation has.
- Empty cs_channel.ini: there is no record for the channels, but channel setup still reads record 1, which does not exist. Give at least one record.

Two files are needed as soon as other constituents are listed in constituents.cs, because they are read for every HRU without a check that they were found:

- cs_irrigation is read for every HRU during setup.
- cs_uptake is read every day for every HRU with a growing plant.

Provide both whenever constituents are simulated, with zero values if there is nothing to add.

cs_hru.ini

Initial constituents in HRU soils. Lines 1 to 5 are skipped. Each record then takes three lines:

Line of record	Content
1	Record name

Line of record	Content
2	Soil values: one dissolved concentration (mg/L) per constituent, then one sorbed concentration (mg/kg of soil) per constituent
3	Plant values: the same number of values, read but not used

An HRU uses the record named in the `cs` column of its `soil_plant.ini` record; if the name is not found, the first record is used. `soil_plant.ini` has the `cs` column only when `event` in `codes.bsn` is not 0. With `event` at 0, the column is not read and every HRU uses the first record of `cs_hru.ini`. Every soil layer starts with the same values.

cs_aqu.ini

Initial constituents in aquifers. Lines 1 to 3 are skipped. Each later line is one record: name, one dissolved concentration (mg/L) per constituent, then one sorbed concentration (mg/kg) per constituent. The sorbed mass is calculated with an assumed aquifer bulk density of 2000 kg/m³.

Aquifers find their record through `initial.aqu_cs` (title, header, then rows with `name`, `pest`, `path`, `hmet`, `salt`, `cs`). Aquifer n uses row n of `initial.aqu_cs`, whatever the row's name, and the `cs` column of that row names a record in `cs_aqu.ini`. See salt inputs for the same link used by salts.

cs_channel.ini

Initial concentrations in channel water. Lines 1 and 2 are skipped. Each later line is one record: name and one concentration (mg/L) per constituent. Channels find their record through `initial.cha_cs` as described for `salt_channel.ini`, using its `cs` column.

cs_res

Constituent parameters and initial concentrations for reservoirs and wetlands. Lines 1 to 15 are skipped (two title lines, twelve comment lines and a header). The file is ignored when the reservoir nutrient file in `file.cio` is `null`. Each later line is one record:

#	Column	Units	Default	Description
1	`name`	-	-	Record name, named in the `cs` column of `reservoir.res_cs` or `wetland.wet_cs`.
2	`v_seo4`	m/day	0.001	Settling velocity of selenate.
3	`v_seo3`	m/day	0.001	Settling velocity of selenite.
4	`v_born`	m/day	0.001	Settling velocity of boron.
5	`k_seo4`	1/day	0.05	First-order loss rate of selenate.
6	`k_seo3`	1/day	0.03	First-order loss rate of selenite.
7	`k_born`	1/day	0.0	First-order loss rate of boron.
8	`theta_seo4`	-	1.08	Temperature factor for the selenate rate.
9	`theta_seo3`	-	1.08	Temperature factor for the selenite rate.
10	`theta_born`	-	1.08	Temperature factor for the boron rate.
11	`c_seo4`	mg/L	0.0	Initial selenate concentration.
12	`c_seo3`	mg/L	0.0	Initial selenite concentration.
13	`c_born`	mg/L	0.0	Initial boron concentration.

All 13 values must be given; a short line takes its remaining values from the next line. A reservoir without a `cs_res` record starts with no constituents and uses the default rates in the table. `reservoir.res_cs` and `wetland.wet_cs` are described under `salt_res`.

cs_irrigation

Concentrations in irrigation water from outside the model. Lines 1 and 2 are skipped. Each later line is one record: name and one concentration (mg/L) per constituent.

An HRU uses the record at the same position as the `cs_hru.ini` record it uses; the name is not used. In 62.0.1 the transfer of constituents with irrigation water is switched off in water allocation, so these concentrations do not change the constituent balance.

fertilizer.frt_cs

Constituent content of each fertilizer. Line 1 is a title and line 2 a header. Then one row per record of `fertilizer.frt`, in the same order and with the same number of rows:

#	Column	Units	Description
1	`name`	-	Fertilizer name. Not checked against `fertilizer.frt`.
2	`seo4`	kg/kg	Selenate per kg of material applied.
3	`seo3`	kg/kg	Selenite per kg of material applied.
4	`boron`	kg/kg	Boron per kg of material applied.

The mass added (kg/ha) is the amount applied (kg/ha) times the value in the table. Constituents are added in the same cases as salts (see `salt_fertilizer.frt`): with manure operations and decision-table `manure` actions whose manure has an `org_min` entry other than the first record of `manure_om.frt`, and with fertilizer operations on HRUs with standing water. For manure, the whole amount goes into the top soil layer.

cs_reactions

Reaction parameters for soils and aquifers, organised in reaction groups and shale formations. Every HRU and every aquifer must have a row.

Section	Content
Line 1, 2	Title and header, skipped
Line 3	Number of parameters n_p and number of reaction groups n_g
next n_p lines	One parameter per line, with one value per reaction group
next line	Header, skipped
next line	Number of shale formations n_s
next n_s lines	Per shale formation: S/Se ratio of the shale, rate constant for autotrophic reduction of oxygen (1/day), rate constant for autotrophic reduction of nitrate (1/day)
next line	Header, skipped
next lines	One line per HRU, in HRU order: HRU number, reaction group, n_s shale fractions (fraction of the HRU area over each shale formation)
next line	Header, skipped (only when the model has aquifers)
next lines	One line per aquifer, in aquifer order: aquifer number, reaction group, n_s shale fractions

The parameter lines are used in this order:

Line	Parameter	Units	Used for
1	Selenium reduction inhibition factor for nitrate	-	soils, aquifers
2	Oxygen concentration in soil water	mg/L	soils
3	Oxygen concentration in groundwater	mg/L	aquifers
4	Sorption coefficient K_d of selenate	L/kg	soils, aquifers

Line	Parameter	Units	Used for
5	Sorption coefficient K_d of selenite	L/kg	soils, aquifers
6	Sorption coefficient K_d of boron	L/kg	soils, aquifers
7	Rate constant for selenate reduction in soil	1/day	soils
8	Rate constant for selenate reduction in groundwater	1/day	aquifers

The first value of the HRU and aquifer lines is not checked; the line position decides. Further parameter lines are read but not used by soils and aquifers. The rate constant for selenite reduction is not read for soils or aquifers and stays 0, so selenite is not reduced there.

cs_uptake

Specified daily constituent uptake by plants. Lines 1 to 3 are skipped. Then one row per plant in `plants.plt`, in the same order: name and one uptake value (kg/ha per day) per constituent. The plant name is not checked. While a plant has roots, its daily uptake of each constituent is split between the soil layers in proportion to the root mass in each layer and is limited to the mass in the layer.

cs_plants_boron

Boron tolerance of plants. Line 1 is a title and line 2 is skipped, line 3 is a switch (0 = off, 1 = on), lines 4 to 7 are skipped, and then one row per plant in `plants.plt`: name, threshold a and slope b. In 62.0.1 the values are read but no routine uses them, so boron has no effect on plant growth.

cs_urban

Constituent content of sediment washed from impervious areas. Lines 1 and 2 are skipped. Each later line is an urban type name from `urban.urb` and one concentration (mg per kg of sediment) per constituent. In 62.0.1 the values are read but not used.

cs_streamobs

A list of channels for daily concentration and load output to `cs_streamobs_output`: a title line, the number of channels, then one channel number per line. In 62.0.1 the switch that would read it is never set, so the file is not read and the output is not written.

Example

`constituents.cs` for a run with one pesticide, the eight salt ions and the three constituents:

```
constituents.cs: constituent list
1
atrazine
0

0

8
so4 ca mg na k cl co3 hco3
3
seo4 seo3 boron
```

There are no pathogens or heavy metals, so their counts are 0 and each is followed by an empty names line.

A matching `cs_aqu.ini` record with 0.005 mg/L selenate in groundwater and 0.2 mg/kg sorbed selenate, values illustrative:

```
cs_aqu.ini: initial constituents in aquifers
seo4 seo3 boron: dissolved (mg/L), then sorbed (mg/kg)
name          seo4     seo3    boron   seo4_s   seo3_s  boron_s
aqu_cs1      0.005    0.001    0.100    0.200    0.050    1.000
```

Related files

- salt inputs: the salt ions listed in `constituents.cs`, and the link files `initial.aqu_cs`, `initial.cha_cs`, `reservoir.res_cs` and `wetland.wet_cs`.
- `file.cio`: names `constituents.cs`.
- `soil_plant.ini`: names the `cs_hru.ini` record for each HRU.
- `codes.bsn`: `event` decides whether `soil_plant.ini` has the `cs` column.
- `fertilizer.frt`: sets the row order of `fertilizer.frt_cs`.
- `pesticide.pes`, `pathogens.pth`, metals: the other groups in `constituents.cs`.
- `cs_atmo.cli` and `cs_recall.rec`: deposition and point-source loads.
- gwflow inputs: constituents become groundwater solutes when gwflow solute transport is on.

Sources

- Master list: `constit_db_read.f90`; type `cs_db_type` in `constituent_mass_module.f90`
- Readers: `cs_hru_read.f90`, `cs_aqu_read.f90`, `cs_cha_read.f90`, `res_read_csdb.f90`, `cs_irr_read.f90`, `cs_fert_read.f90`, `cs_reactions_read.f90`, `cs_uptake_read.f90`, `cs_plant_read.f90`, `cs_urban_read.f90`
- Link files and initialisation: `soil_plant_init.f90`, `cs_hru_init.f90`, `aqu_read_init_cs.f90`, `sd_channel_read.f90`, `sd_hydsed_init.f90`, `res_read_salt_cs.f90`
- Processes: `cs_fert.f90`, `cs_fert_wet.f90`, `cs_irrig.f90`, `cs_uptake.f90` (called from `pl_biomass_gro.f90`), `cs_sorb_hru.f90`, `cs_sorb_aqu.f90`, `se_reactions_soil.f90`, `se_reactions_aquifer.f90`, `res_cs.f90`
- Types and variables: `cs_module.f90`, `cs_data_module.f90`, `res_cs_module.f90`

File and column	Fortran variable
`constituents.cs` counts	`cs_db%num_pests`, `%num_paths`, `%num_metals`, `%num_salts`, `%num_cs`
`constituents.cs` names	`cs_db%pests`, `%paths`, `%metals`, `%salts`, `%cs`
`cs_hru.ini` soil and plant lines	`cs_soil_ini%soil`, `cs_soil_ini%plt`
`cs_aqu.ini`	`cs_aqu_ini%aqu`
`cs_channel.ini`	`cs_cha_ini%conc`
`cs_res`	`res_cs_data` (type `reservoir_cs_data`)
`cs_irrigation`	`cs_water_irr%water`, copied to `cs_irr%csc`
`fertilizer.frt_cs`	`fert_cs` (type `fert_db_cs`)
`cs_reactions` parameter lines 1 to 8	`rct(1:8, group)`, copied to `se_ino3`, `oxy_soil`, `oxy_aqu`, `kd_seo4`, `kd_seo3`, `kd_born`, `kseo4` of `cs_rct_soil` and `cs_rct_aqu`
`cs_reactions` shale lines	`rct_shale(:,1:3)`, copied to `sseratio`, `ko2a`, `kno3a`
`cs_uptake`	`cs_uptake_kg`
`cs_plants_boron`	`bor_tol_sim`, `bor_stress_a`, `bor_stress_b`
`cs_urban`	`cs_urban_conc`
`cs_streamobs`	`cs_str_obs`